

Isotropic Systems

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In this paper we introduce a combinatorial and algebraic structure called isotropic system. Some isotropic systems are associated to pairs of dual binary matroids, the sum-decomposable systems, and the other ones are associated to 4-regular graphs, the graphic systems, but there are isotropic systems which are neither graphic nor binary. The introduction of the paper describes the applications of isotropic systems that we have found at present and which will be developed in subsequent papers.

1. INTRODUCTION

Isotropic systems are algebraic structures which unify some properties shared by 4-regular graphs and pairs of dual binary matroids. More precisely there is a subclass of isotropic systems associated to 4-regular graphs, called graphic systems, and a subclass of isotropic systems associated to pairs of dual binary matroids, called sum-decomposable systems, but there are also isotropic systems which are neither graphic nor sum-decomposable.

Isotropic systems can be applied in diversified situations which cannot be covered in a single article. This introduction presents briefly these applications as a motivation, and the remainder of the paper develops some basic properties of isotropic systems.

Our initial motivation when introducing isotropic systems was an attempt to unify the properties of the Martin polynomial $m(G, o; \zeta)$ of a 4-regular graph provided with an eulerian orientation o (i.e. $d^+(v) = d^-(v)$ for every vertex v) and those of the symmetric Tutte-polynomial $t(M; \zeta, \zeta)$ of a binary matroid M . Let G be a 4-regular graph. An eulerian decomposition of G is a set D of its closed trails such that each edge of G belongs to precisely one closed trail in D . If G is connected and provided with an eulerian orientation o then its Martin polynomial is $m(G, o; \zeta) = \sum (\zeta - 1)^{|D|-1}$, [10, 19] summed over the eulerian decompositions D whose closed trails are coherently oriented by o . An interesting case is when G is a plane graph and o is an alternate orientation (i.e. when turning around a vertex v of G we meet alternatively edges directed from v and edges directed towards v). We colour the faces delimited by G in black and white in such a way that every pair of faces sharing a common edge on their boundaries get distinct colours. Putting a black (white) central point inside each black (white) face, and creating for each vertex v of G a black (white) edge traversing v and joining the centres of the black (white) faces incident to v , we construct a black (white) plane graph BG (WG). These graphs are dual from each other, and P. Martin [10] proved that $m(G, o; \zeta) = t(M; \zeta, \zeta)$, the evaluation of the Tutte-polynomial of the cycle-matroid M of either BG or WG for its two variables equal to ζ .

If S is an isotropic system then we can define orientations of S and a polynomial $m(S, o'; \zeta)$ attached to S and an orientation o' of S . If S is induced by a 4-regular graph G then the orientations of S correspond to the eulerian orientations of G and we have the equality $m(S, o'; \zeta) = m(G, o; \zeta)$ for every two corresponding orientations o' of S and o of G . If S is induced by a pair of dual binary matroids M and M^* there exists an orientation o' of S such that $m(S, o'; \zeta) = t(M; \zeta, \zeta) = t(M^*; \zeta, \zeta)$. We can recover Martin's theorem as a consequence, but more fundamentally the unification of the Tutte-polynomial and the Martin-polynomial suggests to carry over isotropic systems results which are known for one of these polynomials. For example, P. Martin [10] proved that $|m(G, o; -1)|$ is a power of 2, and Read and Rosenstiehl [11] proved the same thing for $|t(M; -1, -1)|$ in the case

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of a binary matroid M . This is more generally true for $|m(S, o; -1)|$. M. Las Vergnas [9] proved that $m(G, o; 3) = i|m(G, o; -1)|$ with an odd integer i and conjectured $\iota(M; 3, 3) = i|\iota(M; -1, -1)|$ for a binary matroid M . It can be proved that $m(S, o; 3) = i|m(S, o; -1)|$ for an isotropic system S and an orientation o [14].

A second application of isotropic systems is relative to the recognition of circle graphs. A circle graph is defined by considering a finite set of chords of a circle, associating a vertex to each chord and defining an edge between two vertices if the corresponding chords intersect. Let us define the local complementation of a simple graph F at a vertex v as the operation which replaces the subgraph induced on $N(v) = \{w : vw \in E(F)\}$ by the complementary subgraph. Two graphs are locally equivalent if one of them is obtained from the other by successive local complementations. It is easy to verify that the local complementation of a circle graph at a vertex v gives a new circle graph obtained by reversing an arc of circle delimited by the chord corresponding to v . It can be shown that every isotropic system is associated to a class of locally equivalent graphs. Moreover the isotropic system is graphic if and only if these graphs are circle graphs. Following Tutte's line for matroids we define a similar notion of connectivity for isotropic systems [15], and it is possible to state a theorem similar to Tutte's 'wheel and whirl theorem' [12]. It allows one to derive a polynomial algorithm for recognizing graphic systems—and consequently circle graphs [16]—by following a method similar to Inukai and Weinberg's one for graphic matroids [6].

The preceding development shows that an isotropic system can be identified to a class of locally equivalent graphs. It appears that local complementations give new insight in the theory of graph decompositions recently developed by W. Cunningham [4]. The composition of two graphs G_1 and G_2 is defined if $V(G_1) \cap V(G_2)$ contains a single vertex v ; it is the graph G defined by $V(G) = V(G_1 \setminus v) \cup V(G_2 \setminus v)$, $E(G) = E(G_1 \setminus v) \cup E(G_2 \setminus v) \cup \{v_1 v_2 : v_1 v \in E(G_1), vv_2 \in E(G_2)\}$ where $H \setminus v$ denotes the subgraph obtained by deleting a vertex v of a graph H . If $|V(G_1)|, |V(G_2)| \geq 3$ then G is said to be decomposable and $\{V(G_1) - v, V(G_2) - v\}$ is called a split of G . The basic, and elementary, property is that locally equivalent graphs have the same splits. This fact can be extended in the general case of digraphs considered in [4], but now the local complementation of a digraph G at a vertex v is the digraph G' defined by $V(G') = V(G)$, $E(G') = E(G) \Delta T(v)$ where $T(v) = \{(x, y) : x \text{ and } y \in V(G) - v, (x, v) \text{ and } (v, y) \in E(G)\}$. This kind of local complementation allows one to derive a polynomial algorithm for decomposing digraphs of order n with a time-complexity $O(n^3)$, improving the $O(n^4)$ complexity of Cunningham's algorithm [17].

Let us consider again the case of simple graphs. A graph G is prime if it is not decomposable and of order ≥ 3 . The analog of Tutte's 'wheel and whirl theorem' stated in terms of graph decompositions says that every prime G of order > 5 has an elementary i -minor which is prime, where an elementary i -minor is obtained by deleting some vertex to a graph locally equivalent to G . It is easy to verify that every prime graph of order 5 is locally equivalent to the cycle of length 5. Therefore every prime graph of order > 5 contains the 5-cycle as an i -minor (an i -minor is obtained by taking a succession of elementary i -minors).

On the opposite side it can be proved that a graph does not contain the 5-cycle as an i -minor if and only if it is totally decomposable, where a totally decomposable graph is either a graph of order 3 or a graph which is decomposable into two totally decomposable graphs. The isotropic systems associated to totally decomposable graphs are graphic and they play a role similar to series-parallel matroids. The exclusion of the 5-cycle as an i -minor is similar to the characterization of series-parallel matroids by excluding K_4 as a minor.

Finally, and surprisingly, we note that the class of the totally decomposable graphs is equal to the class of the distance-hereditary graphs, where a graph is said to be distance-hereditary if every chordless path is geodesic. This result is easily derived from Bandelt and Mulder's recent characterizations of distance-hereditary graphs [1] but it gives also some new insight on this class of graphs [18].

2. CONVENTIONS

We specify our conventions in matroid theory and graph theory. For the background the reader can refer to [2] and [13].

A 1-element set $\{v\}$ will often be simply denoted v .

If V is a finite set then its set of subsets, $\mathcal{P}(V)$, will be considered as a vector space over $\text{GF}(2)$ with the addition $(P, Q) \rightarrow P + Q = \text{symmetric difference of } P \text{ and } Q$. We consider also on $\mathcal{P}(V)$ the bilinear form $(P, Q) \rightarrow PQ = |P \cap Q| \pmod{2}$. If N is a subspace of $\mathcal{P}(V)$, $P \subseteq V$ and $v \in V$, we define the subspaces

$$\begin{aligned} N \times P &= \{W \in N : W \subseteq P\}, \\ N \setminus v &= \{P \subseteq V - v : P \in N\} = N \times (V - v), \\ N/v &= \{P \subseteq V - v : P \in N \text{ or } P + v \in N\}. \end{aligned}$$

A *binary matroid* (on the set V) will be defined throughout this paper as a pair $M = (N, V)$ with a finite set V and a subspace $N \subseteq \mathcal{P}(V)$. The *circuits* of M are the minimal nonempty sets in N .

The *rank* of a subset $Q \subseteq V$ is given by the formula $r(Q) = |Q| - \dim(N \times Q)$. The rank of M is $r(M) = r(V) = |V| - \dim(N)$. The *dual matroid* of M is $M^* = (N^\perp, V)$ where $N^\perp = \{P \in \mathcal{P}(V) : PQ = 0 \text{ for every } Q \in N\}$ is the orthogonal subspace of N in $\mathcal{P}(V)$. To *delete* (*contract*) an element $v \in V$ in M is to consider the binary matroid $M \setminus v = (N \setminus v, V - v)$ ($M/v = (N/v, V - v)$). A *minor* of M is obtained from M by a sequence of contractions and deletions.

A graph G will be finite, nonoriented, with possible loops and/or multiple edges. The edge-set of G is denoted by $E(G)$, and the vertex-set by $V(G)$. A *cycle* of G is a subset of $E(G)$ defining a subgraph with even degrees. The set of the cycles, denoted $\mathcal{Z}(G)$ is a subspace of $\mathcal{P}(E(G))$ whose dimension is equal to $|E(G)| - |V(G)| + k(G)$ where $k(G)$ is the number of components of G . The *coboundary operator* of G is the linear mapping $\delta : \mathcal{P}(V(G)) \rightarrow \mathcal{P}(E(G))$ defined by $\delta P = \{\text{edges of } G \text{ with one end in } P \text{ and the other end in } V(G) \setminus P\}$. A *cocycle* of G is a subset of $E(G)$ which is equal to some δP , $P \subseteq V(G)$. The set of the cocycles, denoted by $\mathcal{C}(G)$, is a subspace of $\mathcal{P}(E(G))$ whose dimension is equal to $|V(G)| - k(G)$. The *matroid of* G is $M(G) = (\mathcal{Z}(G), E(G))$; its dual matroid is $M(G)^* = (\mathcal{C}(G), E(G))$. A matroid M is said to be *graphic* (*cographic*) if there exists a graph G such that $M = M(G)$ ($M = M(G)^*$).

3. DEFINITION OF AN ISOTROPIC SYSTEM

Let K be a 2-dimensional vector space over $\text{GF}(2)$. We provide K with the bilinear form $(x, y) \rightarrow xy$ such that $xy = 1 \Leftrightarrow 0 \neq x \neq y \neq 0$. For every finite set V we shall consider that the vector space K^V is provided with the bilinear form $(A, B) \rightarrow AB = \sum (A(v)B(v) : v \in V)$. Therefore two vectors A and B will be orthogonal in K^V if and only if $|\{v \in V : 0 \neq A(v) \neq B(v) \neq 0\}|$ is even. In particular every vector of K^V is orthogonal to itself. A subspace L of K^V is *totally isotropic* if every two vectors of L are orthogonal.

An *isotropic system* is a pair (L, V) with a finite set V and a totally isotropic subspace L of K^V with a dimension equal to $|V|$.

For every $v \in V$ and every $x \in K$ let $s(v, x)$ be the vector of K^V defined by $s(v, x)(v) = x$ and $s(v, x)(w) = 0$ for $w \in V - v$. If $(P_v : v \in V)$ is a family of linear endomorphisms of K then the linear endomorphism P of K^V which maps $s(v, x)$ onto $s(v, P_v(x))$ for each $(v, x) \in V \times K$ is said to be *induced* by $(P_v : v \in V)$. Two isotropic systems $S = (L, V)$ and $S' = (L', V')$ are *strongly isomorphic* if $V = V'$ and there exists a family $(P_v : v \in V)$ of linear endomorphisms of K such that $L' = P(L)$. The family $(P_v : v \in V)$ is called a *strong isomorphism* from S into S' . We notice that a linear endomorphism P_v of K is characterized by: $P_v(0) = 0$ and the restriction of P_v to $K - 0$ is bijective.

4. SUM-DECOMPOSABLE SYSTEMS

Let V be a finite set. For $A \in K^V$ and $P \subseteq V$ we denote by AP the vector of K^V defined by $AP(v) = A(v)$ if $v \in P$ and $AP(v) = 0$ if $v \notin P$. We set $\hat{A} = \{AP : P \subseteq V\}$. It is easy to verify that $\hat{A}$ is a totally isotropic subspace of K^V . The vector A is said to be *complete* if $A(v) \neq 0$ for every $v \in V$. This occurs if and only if the mapping $P \rightarrow AP$ from $\mathcal{P}(V)$ into $\hat{A}$ is linear and bijective. Two vectors $A, B \in K^V$ are said to be *supplementary* if $0 \neq A(v) \neq B(v) \neq 0$ for every $v \in V$. This occurs if and only if $\hat{A}$ and $\hat{B}$ are supplementary subspaces of K^V .

(4.1) *Let A and B be supplementary vectors of K^V , and $P, Q \in \mathcal{P}(V)$. Then $APBQ = PQ$.*

PROOF. For $v \in V$ the product $AP(v)BQ(v)$ is null if $v \notin P \cap Q$ otherwise it is equal to $A(v)B(v)$. In the last case we have $0 \neq B(v) \neq A(v) \neq 0$ because A and B are supplementary. Therefore $APBQ = \sum (1 : v \in P \cap Q) = |P \cap Q| \pmod{2}$.

For a subspace $N \subseteq \mathcal{P}(V)$ and a complete vector $A \in K^V$, we set $AN = \{AP : P \in N\}$. Since $P \rightarrow AP$ is linear and bijective from $\mathcal{P}(V)$ onto $\hat{A}$, AN is a subspace of $\hat{A}$ and $\dim(AN) = \dim(N)$.

(4.2) *If $M = (N, V)$ and $M^* = (N^\perp, V)$ are dual binary matroids and A and B are supplementary vectors of K^V , then $(AN + BN^\perp, V)$ is an isotropic system.*

PROOF. Let us consider $X_1, X_2 \in AN + BN^\perp$. There exist $P_i \in N$ and $Q_i \in N^\perp$ such that $X_i = AP_i + BQ_i$ ($i = 1, 2$). This yields

$$X_1X_2 = AP_1AP_2 + BQ_1BQ_2 + AP_1BQ_2 + AP_2BQ_1.$$

The product AP_1AP_2 is null because AP_1 and AP_2 belong to the totally isotropic subspace $\hat{A}$. Similarly $BQ_1BQ_2 = 0$. Applying (4.1) we obtain thus $X_1X_2 = P_1Q_2 + P_2Q_1$. But $P_1Q_2 = 0$ because $P_1 \in N$ and $Q_2 \in N^\perp$. Similarly $P_2Q_1 = 0$. Therefore $AN + BN^\perp$ is a totally isotropic subspace of K^V .

We have $\dim(AN + BN^\perp) = \dim(AN) + \dim(BN^\perp)$ because $AN \subseteq \hat{A}$, $BN^\perp \subseteq \hat{B}$, $\hat{A}$ and $\hat{B}$ are supplementary subspaces of K^V . But $\dim(AN) = \dim(N)$ and $\dim(BN^\perp) = \dim(N^\perp)$. Therefore $\dim(AN + BN^\perp) = |V|$.

An isotropic system $(AN + BN^\perp, V)$ which is so associated to a pair of dual binary matroids $M = (N, V)$ and $M^* = (N^\perp, V)$ will be called a *sum-decomposable system*. It will be proved in [3] that some isotropic systems are not sum-decomposable.

(4.3) *Let $S = (AN + BN^\perp, V)$ be a sum-decomposable system associated to a pair of dual binary matroids $M = (N, V)$ and $M^* = (N^\perp, V)$. Another isotropic system $S' = (L', V)$ is strongly isomorphic to S if and only if it is also sum-decomposable and associated to M and M^* .*

PROOF. If $S' = (A'N + B'N^\perp, V)$ let us consider the family $(P_v : v \in V)$ of linear endomorphisms of K defined by $P_v(A(v)) = A'(v)$ and $P_v(B(v)) = B'(v)$. Let P be the induced endomorphism of K^V . The images by P of AN and $BN^\perp$ are respectively $A'N$ and $B'N^\perp$. Therefore the image of $AN + BN^\perp$ is $A'N + B'N^\perp$ and so $(P_v : v \in V)$ is a strong isomorphism of S into S' . Conversely if the strong isomorphism $(P_v : v \in V)$ is given then we verify that $S' = (A'N + B'N^\perp, V)$, where $A'(B')$ is the image of $A(B)$ by P .

So a pair of dual binary matroids determines a sum-decomposable system up to a strong isomorphism.

5. GRAPHIC SYSTEMS

A *coded graph* is a pair $(G, (T_v: v \in V))$ defined in the following way. G is a 4-regular graph, and each edge e of G is viewed as a pair $\{h', h''\}$ of *half-edges* having one end each, in such a way that the ends of e are the ends of h' and h'' . If v is a vertex of G then hv denotes the set of the four half-edges with an end equal to v , and if C is a subset of edges of G then hC denotes the set of the half-edges of the edges of C . Therefore C is a cycle if and only if $|hv \cap hC| = 0, 2$ or 4 for every vertex v of C . A *transition* at a vertex v of G is a pair of distinct half-edges of hv . We define a mapping T_v which maps the set of the six transitions at v into $K - 0$ in such a way that two transitions have the same image if and only if they are either equal or disjoint. For each transition t at v , $T_v(t)$ is called the *code* of t . We extend the definition of T_v by setting $T_v(\emptyset) = T_v(hv) = 0$. So T_v is now defined on the set $\mathcal{P}_2(hv)$ consisting of the subsets of hv which have an even cardinality. The set $\mathcal{P}_2(hv)$ is a subspace of $\mathcal{P}(hv)$, and it is clear that $T_v: \mathcal{P}_2(hv) \rightarrow K$ is linear. We define such a T_v for each vertex v of G , and we call $(T_v: v \in V)$ a *transition coding*. Finally we define for each cycle C of G a vector $T(C) \in K^V$, called the *tangent* to C , by $T(C)(v) = T(hv \cap hC)$ for every vertex v . The mapping $T: \mathcal{Z}(G) \rightarrow K^V$ is the *tangent mapping* over $(G, (T_v: v \in V))$; it is clearly a linear mapping.

Informally we can imagine that G is drawn in the plane with some possible crossings of edges (in particular if G is not planar). At each vertex v of G let us draw a little line separating two disjoint transitions t_1 and t_2 at v . Let us imagine an orthogonal line separating two other disjoint transitions t_3 and t_4 at v (the orthogonal lines are not drawn in Figure 1(a) because there is a single pair of disjoint transitions $\{t_3, t_4\} \neq \{t_1, t_2\}$ which can be separated by a line at v). Finally there remains two disjoint crossing transitions t_5 and t_6 which cannot be separated by a line. Let us suppose that $K = \{0, x, y, z\}$. We obtain a transition coding $(G, (T_v: v \in V))$ by setting $T_v(t_1) = T_v(t_2) = x$, $T_v(t_3) = T_v(t_4) = y$, $T_v(t_5) = T_v(t_6) = z$, $T_v(\emptyset) = T_v(hv) = 0$. If C is an elementary cycle such that every pair of successive edges of C defines a noncrossing transition, then $T(C)(v) = 0$ if v is not incident to C otherwise $T(C)(v)$ is the value attached to the line 'tangent' to C at v .

For example if we consider in Figure 1(a) the elementary cycle C_1 defined by the vertex-sequence (A, B, C, D, E), the corresponding tangent $T(C_1)$ is given in Figure 1b. The

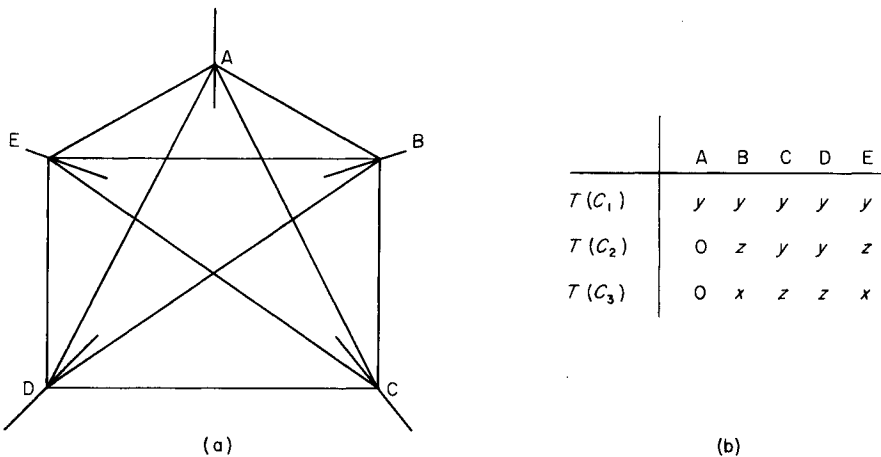


FIGURE 1

mapping $T: \mathcal{Z}(G) \rightarrow K^V$ generalizes this notion of tangent to every cycle of G . Figure 1(b) also gives $T(C_2)$ for the elementary cycle C_2 described by the vertex-sequence (B, C, D, E) and $T(C_3)$ for C_3 equal to the union of the triangles (A, B, E) and (A, C, D).

(5.1) *Let $(G, (T_v: v \in V))$ be a coded graph, and let $T: \mathcal{Z}(G) \rightarrow K^V$ be the associated tangent mapping. A subset C of $E(G)$ is in the kernel of T if and only if C is the edge-set of a union of components of G . If L is the image of T , then (L, V) is an isotropic system.*

PROOF. If $C \in \text{Ker}(T)$ then we have $T_v(hC \cap hv) = 0$ for every vertex v , so that $hC \cap hv = \emptyset$ or hv . Let us consider the function $f: V \rightarrow \text{GF}(2)$ defined by $f(v) = 0$ if $hC \cap hv = \emptyset$ and $f(v) = 1$ if $hC \cap hv = hv$. For an edge e of G with the ends v' and v'' , we have $f(v) = f(v'') = 1$ if $e \in C$ and $f(v') = f(v'') = 0$ if $e \notin C$. Therefore f is constant on $V(\Gamma)$ for each component Γ of G , and $E(\Gamma) \subseteq C$ if $f = 1$ on $V(\Gamma)$, $E(\Gamma) \subseteq E(G) \setminus C$ if $f = 0$ on $V(\Gamma)$. Therefore $C \in \text{Ker}(T)$ implies that C is the edge-set of a union of components of G . The converse is obvious.

The preceding property implies that $\dim(\text{Ker}(T)) = k(G)$, where $k(G)$ denotes the number of components of G . But we have

$$\dim(\text{Ker}(T)) + \dim(\text{Im}(T)) = \dim(\mathcal{Z}(G)) = |E(G)| - |V| + k(G).$$

Moreover $|E(G)| = 2|V|$ because G is 4-regular. This implies $\dim(\text{Im}(T)) = |V|$. In order to prove that (L, V) is an isotropic system it remains to verify that every two vectors $T(C_1)$ and $T(C_2)$ of L are orthogonal. We have

$$T(C_1) T(C_2) = |\{v \in V: 0 \neq T(C_1)(v) \neq T(C_2)(v) \neq 0\}| \pmod{2}.$$

The values $T(C_1)(v)$ and $T(C_2)(v)$ are nonnull if and only if $hC_1 \cap hv$ is a transition t_1 at v and $hC_2 \cap hv$ is a transition t_2 at v . The values $T(C_1)(v)$ and $T(C_2)(v)$ are distinct if and only if t_1 and t_2 are neither equal nor disjoint, which is equivalent to $|t_1 \cup t_2| = 3$. This implies that $\{v \in V: 0 \neq T(C_1)(v) \neq T(C_2)(v) \neq 0\}$ is the set of the vertices with an odd degree in the subgraph of G defined by the edge-set $C_1 \cup C_2$. Therefore $T(C_1)T(C_2) = 0$.

An isotropic system $S = (L, V)$ which is associated to a coded graph $(G, (T_v: v \in V))$ by setting $L = \{T(C): C \in \mathcal{Z}(G)\}$ will be called a *graphic system*. It will be proved in [3] that some isotropic systems are not graphic.

(5.2) *Let S be a graphic system associated to a coded graph $(G, (T_v: v \in V))$. Another isotropic system S' defined on V is strongly isomorphic to S if and only if it is graphic and associated to a coded graph $(G, (T'_v: v \in V))$ with the same underlying 4-regular graph G .*

PROOF. If S' is associated to $(G, (T'_v: v \in V))$ then let us consider the family $(P_v: v \in V)$ of linear endomorphisms of K defined by $P_v(T_v(t)) = T'_v(t)$ for every vertex v of G and every transition t at v . Let P be the induced endomorphism of K^V . If we consider the tangent mappings associated to $(G, (T_v: v \in V))$ and $(G, (T'_v: v \in V))$, we have for every cycle C and every vertex v

$$T'(C)(v) = T'_v(hC \cap hv) = P_v(T_v(hC \cap hv)) = P_v(T(C)(v)).$$

Therefore the image of $T(C)$ by P is equal to $T'(C)$ for every cycle C of G , and so $(P_v: v \in V)$ is a strong isomorphism from S into S' . Conversely if the strong isomorphism $(P_v: v \in V)$ is given we verify that S' is associated to $(G, (T'_v: v \in V))$, where the transition-coding $(T'_v: v \in V)$ is defined by $P_v(T_v(t)) = T'_v(t)$ for every vertex v and every transition at v .

So a 4-regular graph determines a graphic system up to a strong isomorphism.

6. EULERIAN DECOMPOSITIONS AND TOUCH-GRAPHS

In this section we show that the isotropic system associated to a coded graph reflects the structure of its eulerian decompositions. We give first a precise definition of a closed trail since the usual one is inadequate when loops are involved.

Let $(G, (T_v : v \in V))$ be a coded graph. A *closed trail* of G is a sequence of pairwise distinct half-edges, $U = (h_0, h_1, h_2, \dots, h_{2l-2}, h_{2l-1})$, $l > 0$, such that $\{h_{2i}, h_{2i+1}\}$ is a transition and $\{h_{2i+1}, h_{2i+2}\}$ is an edge for $i = 0, 1, \dots, l-1$ and the convention $2l+2 = 0$ if $i = l-1$. The vertex-sequence of U is $(v_0, v_2, \dots, v_{2l-2})$ where v_{2i} is the vertex at which the transition $\{h_{2i}, h_{2i+1}\}$ is defined. The half-edges, edges, transitions and vertices of U are defined in the obvious way. We notice that the set of edges of U is a cycle of G . We convince that after cyclically permuting the sequence U and/or reversing it we obtain the same closed trail.

If G is constituted of a single vertex and two loops $\{h'_1, h''_1\}$ and $\{h'_2, h''_2\}$ then there are two closed trails containing these loops, $(h'_1, h''_1, h'_2, h''_2)$ and $(h'_1, h''_2, h'_2, h''_1)$. The usual definition of a closed trail ([2] 1.6 and 1.7) does not allow this distinction.

An *eulerian decomposition* of G is a set D of closed trails such that each edge of G belongs to one and only one closed trail in D . A *transition of D* is a transition of a closed trail in D . A *system of transitions* of G is a set of transitions which partition the half-edge set of G . Eulerian decompositions and systems of transitions have been introduced by A. Kotzig [8] who called them respectively ξ -decompositions and δ -systems. It is clear that the set of the transitions of an eulerian decomposition is a system of transitions, and that conversely every system of transitions is the set of the transitions of one and only one eulerian decomposition.

Let A be a complete vector of K^V . A *transition induced by A* is a transition t at a vertex v such that $T_v(t) = A(v)$. The *system of transitions induced by A* is the set of all the transitions induced by A ; it is denoted by $S(A)$. The *eulerian decomposition induced by A* is the eulerian decomposition defined by $S(A)$; it is denoted by $D(A)$. It is clear that every system of transitions (eulerian decomposition) is induced by one and only one complete vector of K^V .

We notice that two complete vectors A and B are supplementary if and only if $S(A) \cap S(B) = \emptyset$. It is natural in our context to say that $S(A)$ and $S(B)$ are *disjoint systems of transitions*, however in other related works, $S(A)$ and $S(B)$ are called 'compatible systems of transitions' (Sabidussi, Fleischner [5]).

Finally we associate to each eulerian decomposition D a graph $\text{Tch}(D)$ which we call the *touch-graph* of D . The vertex-set of $\text{Tch}(D)$ is equal to D and the edge-set of $\text{Tch}(D)$ is equal to V . The ends of $v \in V$ considered as an edge of $\text{Tch}(D)$ are the closed trails of D which are incident to v . If A is a complete vector of K^V , we set $\text{Tch}(A) = \text{Tch}(D(A))$, and we say also that $\text{Tch}(A)$ is the touch-graph of A .

(6.1) *Let (L, V) be the isotropic system associated to a coded graph $(G, (T_v \in V))$. Let A be a complete vector of K^V . A subset $P \subseteq V$ is a cocycle of $\text{Tch}(A)$ if and only if $AP \in L$.*

PROOF. Let δ be the coboundary operator of $\text{Tch}(A)$. We prove first

$$(i) \quad T(C) = A\delta U, \quad U \in D(A), \quad C = \{\text{edges of } U\}.$$

If $v \in \delta U$ then there is a transition t at v induced by A which belongs to U when the second transition t' at v induced by A does not belong to U . Therefore $hC \cap hv = t$, and so $T(C)(v) = A(v)$. If $v \notin \delta U$ then either the two transitions t and t' belong to U or none of these transitions belongs to U . Therefore $hC \cap hv = 0$ or hv , and so $T(C)(v) = 0$. Equality (i) is thus proved.

Let C be a cycle of G such that $T(C) = AP$ for some subset $P \subseteq V$. Let us define a set S of transitions induced by A in the following way. For $v \in P$ we have $T(C)(v) = A(v)$, and so $hC \cap hv$ is a transition induced by A ; we put it into S . For $v \notin P$ we have $T(C)(v) = 0$, and so $hC \cap hv = 0$ or hv . If $hC \cap hv = 0$ no transition at v is put into S . If $hC \cap hv = hv$ then the two transitions at v induced by A are put into S . It is clear that S partitions hC . Let $U_1, U_2, \dots, U_k$ be the closed trails of G whose transitions belong to S . These closed trails are in $D(A)$. Each edge of C belongs to exactly one of the U_i s because S partitions hC . Therefore if we denote by C_i the set of the edges of U_i ($i = 1, 2, \dots, k$) then C is the disjoint union of $C_1, C_2, \dots, C_k$. Equality (i) implies

$$T(C_i) = A\delta U_i, \quad i = 1, 2, \dots, k.$$

Since $C = C_1 + C_2 + \dots + C_k$ and the maps T, δ and $P \rightarrow AP$ are linear, we obtain by adding the preceding equalities for $i = 1, 2, \dots, k$

$$T(C) = A\Sigma(\delta U_i; 1 \leq i \leq k)$$

proving the 'if' part with $P = \Sigma(\delta U_i; 1 \leq i \leq k)$.

Conversely let us suppose that P is a cocycle of $\text{Tch}(A)$. There exist closed trails $U_1, U_2, \dots, U_k$ in $D(A)$ such that $P = \Sigma(\delta U_i; 1 \leq i \leq k)$. Let $C = C_1 + C_2 + \dots + C_k$ where C_i is the set of the edges of U_i . Equality (i) and the linearity of the mappings $T, \delta, P \rightarrow AP$ imply $T(C) = AP$, proving the 'only if' part.

The following property will be usefull for the sequel.

(6.2) *For a coded graph $(G, (T_v; v \in V))$ and a complete vector A of K^V , G and $\text{Tch}(A)$ have the same number of components.*

PROOF. Let us consider v' and v'' in V . If v' and v'' considered as vertices of G are the endpoints of an edge $e \in E(G)$ then v' and v'' are vertices incident to the closed trail $U \in D(A)$ which contains e . Therefore v' and v'' considered as edges of $\text{Tch}(A)$ are incident to the vertex U . So if $P \subseteq V$ is a connected subset of vertices in G , it is also a connected subset of edges of $\text{Tch}(A)$.

If v' and v'' considered as edges of $\text{Tch}(A)$ are incident to a same vertex U , then U is a closed trail of G which contains a path joining v' and v'' considered as vertices of G .

Therefore if $P \subseteq V$ is a connected subset of edges in $\text{Tch}(A)$, it is also a connected subset of vertices in G . Since the converse is true, and neither G nor $\text{Tch}(A)$ have isolated vertices, G and $\text{Tch}(A)$ have the same number of components.

7. PROJECTION PARALLEL TO A COMPLETE VECTOR

Let (L, V) be an isotropic system and A be a complete vector of K^V . We set $\text{Int}(A) = \{P \subseteq V: AP \in L\}$. The intersection $L \cap \hat{A}$ is a subspace of $\hat{A}$, and $\text{Int}(A)$ is its inverse image by the linear isomorphism $P \rightarrow AP$ from $\mathcal{P}(V)$ into $\hat{A}$. Therefore $\text{Int}(A)$ is a subspace of $\mathcal{P}(V)$. We define the *rank* of A as $r(A) = \dim(\text{Int}(A)) = \dim(L \cap \hat{A})$.

Let B be a supplementary vector of A . Since $\hat{A}$ and $\hat{B}$ are supplementary subspaces of K^V , each vector $z \in K^V$ has a unique decomposition $z = z_A + z_B$, $z_A \in \hat{A}$, $z_B \in \hat{B}$. We set $\text{Prj}(A) = \{P \subseteq V: z_B = BP \text{ for some } z \in L\}$, and we show below that $\text{Prj}(A)$ does not depend on B . The set $\{z_B: z \in L\}$, the projection of L on $\hat{B}$ parallel to $\hat{A}$ is a subspace of $\hat{B}$, and $\text{Prj}(A)$ is its inverse image by the linear isomorphism $P \rightarrow BP$ from $\mathcal{P}(V)$ into $\hat{B}$. Therefore $\text{Prj}(A)$ is a subspace of $\mathcal{P}(V)$.

$$(7.1) \quad \text{Prj}(A) = \text{Int}(A)^\perp, \text{ the orthogonal subspace of } \text{Int}(A) \text{ in } \mathcal{P}(V).$$

PROOF Let us consider $X \in \text{Prj}(A)$ and $Y \in \text{Int}(A)$. There exists $1 \in L$ such that $1 = 1_A + 1_B$, $1_B = BX$. Since AY and 1_A are in $\hat{A}$ we have

$$0 = (1_A + 1_B)AY = 1_BAY = BXAY = XY \text{ by (4.1).}$$

Therefore $\text{Prj}(A) \subseteq \text{Int}(A)^\perp$.

Let us consider a supplementary subspace Z of $L \cap \hat{A}$ in L . We have $\dim(Z) = |V| - \dim(L \cap \hat{A}) = \dim(\text{Int}(A)^\perp)$. The linear mapping $z \rightarrow z_B$ from Z into $\hat{B}$ is injective because $z_B = 0$ implies $z = z_A$, but $z_A \in \hat{A}$ and $Z \cap \hat{A} = \{0\}$. Therefore

$$\dim(\text{Prj}(A)) \geq \dim\{z_B : z \in Z\} = \dim(Z) = \dim(\text{Int}(A)^\perp).$$

This inequality implies $\text{Prj}(A) = \text{Int}(A)^\perp$.

The projection of (L, V) parallel to A is the binary matroid $M(A) = (\text{Prj}(A), V)$. The preceding property means that $M(A)^* = (\text{Int}(A), V)$.

(7.2) Let (L, V) be the isotropic system attached to a coded graph $(G, (T_v : v \in V))$. Let A be a complete vector of K^V . Then

$$M(A) = M(\text{Tch}(A)),$$

$$r(A) = r(M(A)),$$

$$r(A) = k(A) - k(G)$$

where $k(A)$ denotes the number of the closed trails in the eulerian decomposition $D(A)$, and $k(G)$ is the number of components of G .

PROOF. Proposition (6.1) implies $\mathcal{C}(\text{Tch}(A)) = \text{Int}(A)$. This means $M(\text{Tch}(A))^* = M(A)^*$, and the first equality follows. The second equality can be obtained by a simple computation. Finally the rank of $M(\text{Tch}(A))$ is equal to $k(A) - k(\text{Tch}(A))$ where $k(\text{Tch}(A))$ denotes the number of components of $\text{Tch}(A)$. But (6.2) says that $k(\text{Tch}(A)) = k(G)$, and the last equality follows.

(7.3) Projections of graphic systems are graphic matroids.

8. MINORS

Let $S = (L, V)$ be an isotropic system. We consider an element v of V and the canonical projection $p_v : K^V \rightarrow K^{V-v}$. We set

$$L_x^v = \{A \in L : A(v) = x\} \quad (x \in K),$$

$$L|_x^v = p_v(L_0^v \cup L_x^v) \quad (x \in K - 0).$$

(8.1) $S|_x^v = (L|_x^v, V - v)$ is an isotropic system.

PROOF. The subset $L_0^v \cup L_x^v$ is a subspace of L because the ground field is $\text{GF}(2)$. $L|_x^v$ is a subspace of K^{V-v} because it is the image of $L_0^v \cup L_x^v$ by the linear mapping p_v .

Let us consider $A', B' \in L|_x^v$ and $A, B \in L_0^v \cup L_x^v$ such that $A' = p_v(A)$ and $B' = p_v(B)$. We have $A(v)/B(v) = 0$ because $A(v)$ and $B(v)$ are both equal to 0 or x . This implies $A'B' = \sum (A(w)B(w) : w \neq v) = AB = 0$. Therefore $L|_x^v$ is totally isotropic. It remains to show that $\dim(L|_x^v) = |V - v| = |V| - 1$.

L_0^v is a subspace of L and each nonempty subset L_y^v ($y \in K$) is a coset of L_0^v in L . Since the ground-field is $\text{GF}(2)$, and $L = U(L_x^v : x \in K)$, we have $\dim(L_0^v) = \dim(L) - q = |V| - q$ where 2^q is the number of the nonempty sets in the family $(L_y^v : y \in K)$. The

restriction of p_v over L_0^v is bijective and $p_v(L_0^v) \subseteq L_x^v$; therefore we have $\dim(L_x^v) \geq |V| - q$. Since L_x^v is a totally isotropic subspace of K^{V-v} , its dimension cannot exceed $|V| - 1$. Therefore $q = 0$ is impossible and if $q = 1$ we have actually $\dim(L_x^v) = |V| - 1$. There remains the case where $q = 2$. Let us suppose that $p_v(L_0^v) \cap p_v(L_x^v) \neq \emptyset$, so that we can find $A' \in L_0^v$ and $A'' \in L_x^v$ with $p_v(A') = p_v(A'')$. Then we have $p_v(A' + A'') = 0$ and the vector $A = A' + A''$ is nonnull because $A'(v) + A''(v) = x$. Therefore $A(w) = 0$ for every $w \neq v$ and $A(v) = x$. Since the number of the nonempty subsets in $(L_y^v: y \in K)$ is equal to $2^q = 4$, we can choose an element $y \in K \setminus \{0, x\}$ and a vector $B \in L_y^v$. We have then $AB = A(v)B(v) = xy = 1$, a contradiction with $A, B \in L$. Therefore we have $p_v(L_0^v) \cap p_v(L_x^v) = \emptyset$, and this implies $|L_x^v| = |p_v(L_0^v)| + |p_v(L_x^v)| = |L_0^v| + |L_x^v|$ because the restrictions of p_v over L_0^v and over L_x^v are bijective. But $|L_x^v| = |L_0^v|$ because L_x^v is a coset of L_0^v . Therefore we have $|L_x^v| = 2|L_0^v|$, and so $\dim(L_x^v) = |V| - 1$.

The isotropic system $S|_x^v$ will be called an *elementary minor* of S . A *minor* of S is obtained through a succession of elementary minors. We give now interpretations of minors in the graphic and sum-decomposable cases.

Let $(G, (T_v: v \in V))$ be a coded graph. For $v \in V$ and $x \in K - 0$ we construct a new graph $G|_x^v$ in the following way. Informally we split the vertex v into two vertices of degree 2 according to the transitions coded by x and we 'erase' the resulting vertices of degree 2 in order to obtain a 4-regular graph. Formally let $\{h_1, h_2\}$ and $\{h_3, h_4\}$ be the transitions at v which are coded by x . Let h'_i , $1 \leq i \leq 4$, be the half-edge which is paired with h_i in an edge of G . We delete from G the vertex v and $hv = \{h_1, h_2, h_3, h_4\}$. The graph $G|_x^v$ will be constituted of the vertex-set $V - v$, of the edges of G with no half-edge in hv and of two, one or zero edges obtained by pairing as follows the half-edges of $\{h'_1, h'_2, h'_3, h'_4\} \setminus hv$. If no loop was incident to v then none of the half-edges h'_1, h'_2, h'_3, h'_4 was in hv ; we pair $\{h'_1, h'_2\}$ and $\{h'_3, h'_4\}$. If one loop was incident to v then two of the half-edges h'_1, h'_2, h'_3, h'_4 were in hv ; we pair the two remaining ones. If two loops were incident to v then none of the half-edges h'_1, h'_2, h'_3, h'_4 remains, and we pair nothing.

The coded graph $(G|_x^v, (T_w: w \in V - v))$ is called an *elementary s -minor* of $(G, (T_v: v \in V))$. A *s -minor* of $(G, (T_v: v \in V))$ is obtained through a succession of elementary s -minors. The prefix s in s -minor recalls that the notion is defined with respect to splitting a vertex and is not to be confused with the usual notion of minor of a graph G obtained through a succession of deletions and contractions of edges.

(8.2) If the isotropic system $S = (L, V)$ is induced by the coded graph $(G, (T_v: v \in V))$ then $S|_x^v$ is induced by $(G|_x^v, (T_w: w \in V - v))$.

PROOF. Let us consider the subspace $\mathcal{Z}|_x^v = \{C \in \mathcal{Z}(G): T(C)(v) = 0 \text{ or } x\}$ where T is the tangent mapping associated to $(G, (T_v: v \in V))$. The reader will verify that $C \in \mathcal{Z}|_x^v$ if and only if $hC \setminus hv = hC'$ for some $C' \in \mathcal{Z}(G|_x^v)$. Moreover we have $T(C)(w) = T'(C')(w)$ for every $w \in V - v$, where T' is the tangent mapping associated to $(G|_x^v, (T_w: w \in V - v))$.

(8.3) COROLLARY. *Minors of graphic systems are graphic systems.*

We recall the following relations for a binary matroid $M = (N, V)$ its dual $M^* = (N^\perp, V)$ and an element $v \in V$:

$$M^*/v = (N^\perp/v, V - v) = ((N \setminus v)^\perp, V - v) = (M \setminus v)^*,$$

$$M^* \setminus v = (N^\perp \setminus v, V - v) = ((N/v)^\perp, V - v) = (M/v)^*.$$

(8.4) For a binary system $S = (AN + BN^\perp, V)$ and $v \in V$, we have:

$$S|_{A(v)}^\vee = (A'(N/v) + B'(N^\perp \setminus v), V - v),$$

$$S|_{B(v)}^\vee = (A'(N \setminus v) + B'(N^\perp / v), V - v)$$

where A' and B' are the restrictions of A and B to $V - v$.

PROOF. Let $L = AN + BN^\perp$. We have $X' \in L|_{A(v)}^\vee$ if and only if there exists $X \in L$ such that $X(v) = 0$ or $X(v) = A(v)$. The vector X is decomposable as $X = AP + BQ$, and so $X(v) = 0$ or $A(v)$ is equivalent to $v \notin Q$. This implies $X' = A'P' + B'Q'$ where $P', Q' \subseteq V - v$ with $P' = P \cap (V - v)$, $Q' = Q$. But $P \in N$ and $P' = P \cap (V - v) \Leftrightarrow P' \in N/v$. Similarly $Q \in N^\perp$, $Q' = Q$ and $v \notin Q \Leftrightarrow Q' \in N^\perp \setminus v$. This yields the first equality. The second one is similar.

We shall prove in [3] that the third minor $S|_{C(v)}^\vee$ is not sum-decomposable in general.

9. TRIANGLES OF EULERIAN DECOMPOSITIONS (COMPLETE VECTORS)

Let $(G, (T_v : v \in V))$ be a coded graph, and v be a vertex of G . Two eulerian decompositions of G are *neighbour at v* if their transitions are equal at every vertex $w \neq v$ and distinct at v . Since there are three possible pairs of transitions at v , a given eulerian decomposition D has exactly two neighbours at v , and these neighbours D' and D'' are obviously neighbour from each other. We call $\{D, D', D''\}$ a *triangle of eulerian decompositions at v* .

(9.1) Let t be a triangle of eulerian decompositions at a vertex v of a coded graph $(G, (T_v : v \in V))$. There exists precisely one eulerian decomposition $D \in t$ such that v is not a loop of the touch-graph $\text{Tch}(D)$. If D' and D'' are the two other eulerian decompositions of t , then $\text{Tch}(D')$ and $\text{Tch}(D'')$ are isomorphic to the graph obtained by identifying the ends of v in $\text{Tch}(D)$.

PROOF. Let us consider some $D \in t$. Since the degree of v is 4, v appears precisely one time in the vertex-sequences of two closed trails $U_1, U_2 \in D$ or it appears precisely twice in the vertex-sequence of a closed trail $U \in D$, and in both case v does not appear in another vertex-sequence of a closed trail of D .

Let us consider the first case. The vertex-sequences of U_1 and U_2 can be written as (v, σ_1) and (v, σ_2) where σ_1 and σ_2 are some sequences of vertices. In $\text{Tch}(D)$, U_1 and U_2 are the ends of v , and so v is not a loop of $\text{Tch}(D)$. When changing the transitions of D at v , we do not change the closed trails of D which are not incident to v , and we change $\{U_1, U_2\}$ either into a closed trail U' with a vertex-sequence equal to $(v, \sigma_1, v, \sigma_2)$ for defining an eulerian decomposition D' or into a closed trail U'' with a vertex-sequence equal to $(v, \sigma_1, v, \tilde{\sigma}_2)$ where $\tilde{\sigma}_2$ is obtained by reversing σ_2 for defining an eulerian decomposition D'' . Therefore an edge is incident to $U'(U'')$ in $\text{Tch}(D')$ ($\text{Tch}(D'')$) if and only if it was incident either to U_1 or to U_2 in $\text{Tch}(D)$. Therefore $\text{Tch}(D')$ and $\text{Tch}(D'')$ are isomorphic to $\text{Tch}(D)$ after identifying U_1 and U_2 .

If the second case occurs we can write the vertex-sequence of U as $(v, \sigma_1, v, \sigma_2)$ where σ_1 and σ_2 are some sequences of vertices. It is the same case as above after exchanging the roles played by D and D' .

We shall now generalize the preceding result to isotropic systems. If V is a finite set, we say that two complete vectors of K^V are neighbour at v if they are equal at any vertex $w \neq v$ and distinct at v . Since there are three nonnull elements in K , a given complete vector A has

exactly two neighbours at v , and these neighbours A' and A'' are neighbour from each other. We call $\{A, A', A''\}$ a triangle of complete vectors at v .

If M is a binary matroid on V and if v is not a loop of M , then the *pseudo-contraction* of v is the operation which consists in adding v as a loop in the contracted matroid M/v . Therefore if $M = (N, V)$, the pseudo-contraction of v gives the matroid $(N/v, V)$. If we identify the ends of an edge v which is not a loop of a graph H , we obtain a graph H' such that $M(H)$ gives $M(H')$ by a pseudo-contraction of v .

(9.2) *Let $S = (L, V)$ be an isotropic system. A vertex v of S is not a loop of a projection of S parallel to a complete vector A if and only if there exists a in $L \cap \hat{A}$ such that $a(v) \neq 0$.*

PROOF. Since $M(A)^* = (\text{Int}(A), V)$, v will be a loop of $M(A)$ if and only if $v \notin X$ for every $X \in \text{Int}(A)$. This means that $a(v) = 0$ for every $a \in L \cap \hat{A}$.

(9.3) *Let $S = (L, V)$ be an isotropic system and v be a vertex of S . Let A be a complete vector of K^V and A' be the restriction of A to $V' = V - v$. Let S' be the elementary minor $S|_{A(v)}$. The projection $M(A')$ of S' parallel to A' and the projection $M(A)$ of S parallel to A are such that $M(A') = M(A)\backslash v$.*

REMARK. If we denote by $A\backslash v$ the restriction of A to $V - v$ and observe that $M(A\backslash v)$ is the projection of $S|_{A(v)}$ parallel to $A\backslash v$, then the preceding property is summarized by the formula

$$M(A\backslash v) = M(A)\backslash v.$$

PROOF. Let us consider $\text{Int}(A) = \{X \subseteq V : AX \in L\}$ and $\text{Int}(A') = \{X' \in V' : A'X' \in L'\}$ where $L' = L|_{A(v)}$. A vector a' is in L' if and only if it is the restriction over V' of a vector $a \in L$ verifying $a(v) = 0$ or $a(v) = A(v)$. In particular, if $a' = A'X'$ for some $X' \in \text{Int}(A')$, the vector a is either equal to $A'X'$ or AX where $X = X' + v$. Therefore $X' \in \text{Int}(A')$ if and only if either $X' \in \text{Int}(A)$ or $X' + v \in \text{Int}(A)$. This means that $\text{Int}(A') = \text{Int}(A)\backslash v$. Since $M(A')^* = (\text{Int}(A'), V')$ and $M(A)^* = (\text{Int}(A), V)$, we obtain $M(A') = M(A)\backslash v$.

(9.4) *Let t be a triangle of complete vectors at a vertex v of an isotropic system $S = (L, V)$. There exists precisely one complete vector $A \in t$ such that v is not a loop of the projection $M(A)$. If A' and A'' are the two other complete vectors of t , then $M(A')$ and $M(A'')$ are equal to the binary matroid obtained by the pseudo-contraction of v . Finally we have the equalities $r(A') = r(A'') = r(A) - 1$.*

PROOF. Let us prove by induction on $n = |V|$ that v is not a loop of some projection of S over a complete vector $A \in t$. If $n = 1$ there are precisely four vectors in K^V , and L contains two of these vectors, the null vector and another vector A verifying necessarily $A(v) \neq 0$. The vector A is complete and we have $L = \hat{A}$. It is then a direct consequence of (9.2) that v is not a loop of $M(A)$. If $n > 1$ we consider a vertex $w \neq v$ and we set $V' = V - w$. Let x be the common value at w of the three vectors in t . We consider the minor $S' = S|_x = (L', V')$. The set of the restrictions to V' of the vectors in t is a triangle t' of complete vectors of $K^{V'}$. By induction there exists $A' \in t'$ such that v is not a loop of $M(A')$, the projection of S' parallel to A' . Considering the vector $A \in t$ whose restriction on V' is equal to A' , and applying (9.3) we obtain $M(A') = M(A)\backslash w$. Since v is not a loop of $M(A)\backslash w$, it is not a loop of $M(A)$.

Let us consider another complete vector $A' \neq A$ in t and the subspaces of $\mathcal{P}(V)$, $\text{Int}(A) = \{X \subseteq V : AX \in L\}$ and $\text{Int}(A') = \{X \subseteq V : A'X \in L\}$. Let us suppose that

some $X \in \text{Int}(A')$ contains v . Setting $a' = A'X$ and considering a vector $a \in L \cap \hat{A}$ verifying $a(v) \neq 0$, we have $aa' = a(v)a'(v) + \sum (a(w)a'(w) : w \neq v)$. The summation vanishes because $a(w) = A(w)$ or 0, $a'(w) = A'(w)$ or 0 and $A(w) = A'(w)$. Moreover we have $a(v)a'(v) = 1$ because $a(v) = A(v) \neq A'(v) = a'(v)$. This implies $aa' = 1$, a contradiction with $a, a' \in L$. So we have proved that no X in $\text{Int}(A')$ contains v , or equivalently $\text{Int}(A') = \text{Int}(A') \setminus v$. Since A and A' differ only by their values at v , we have $\text{Int}(A') \setminus v = \text{Int}(A) \setminus v$. Since $M(A')^* = (\text{Int}(A'), V)$ and $M(A)^* = (\text{Int}(A), V)$ (refer to Section 7), we obtain $M(A')^* = (\text{Int}(A) \setminus v, V)$. This means that $M(A')$ is obtained from $M(A)$ by a pseudo-contraction of v .

So the first assertions are proved. The final double equality is an immediate consequence.

The preceding property is basic in the theory of isotropic systems. We shall see in other papers that it is the essential step for extending the Martin–Read–Rosenstiehl theorem to isotropic systems (refer to the introduction), and the following corollary will appear to be equivalent to a result by Jaeger [7] when identifying an isotropic system to a class of locally equivalent graphs.

(9.5) *Corollary.* Let $S = (L, V)$ be an isotropic system. A minor of a projection of S is a projection of a minor of S .

PROOF. Let A be a complete vector of K^V and $v \in V$. We consider the projection $M(A)$ of S parallel to A and we prove the property for the two elementary minors $M(A) \setminus v$ and $M(A)/v$. Following (9.3) we have $M(A) \setminus v = M(A \setminus v)$, the projection of $S|_{A \setminus v}^v$ over $A \setminus v$. If v is a loop of $M(A)$ the property holds because $M(A)/v = M(A) \setminus v$. If v is not a loop of $M(A)$ we consider the triangle t of complete vectors at v which contains A . If A' is another vector of t we know by (9.4) that $M(A')$ is obtained by the pseudo-contraction of v in $M(A)$. Therefore $M(A)/v = M(A') \setminus v$. Using again (9.3) we have $M(A') \setminus v = M(A' \setminus v)$, the projection of $S|_{A' \setminus v}^v$ over $A' \setminus v$. Therefore $M(A)/v = M(A' \setminus v)$.

REFERENCES*

1. H. J. Bandelt and H. M. Mulder, *Distance-hereditary graphs*.
2. J. A. Bondy, U. S. R. Murty, *Graph theory with applications*, North-Holland, New-York, 1981.
3. A. Bouchet, Graphic presentations of isotropic systems, submitted.
4. W. H. Cunningham, Decomposition of directed graphs, *Siam J. Alg. Disc. math.* **3** (1982), 214–228.
5. H. Fleischner, Eulersche Linien und Kreisüberdeckungen die vorgegebene Durchgänge in den Kanten vermeiden, *J. Combin. Theory, Ser. B* **29** (1980), 145–167.
6. T. Inukia and L. Weinberg, Graph realizability of matroids, *Ann. New York Acad. Sci.* **319** (1979), 289–305.
7. F. Jaeger, On some algebraic properties of graphs, in: *Proceedings of the Silver Jubilee Conference on Combinatorics, Waterloo, June 1982*, Academic press, London, 1984, pp. 347–366.
8. A. Kotzig, Eulerian lines in finite 4-valent graphs and their transformations, in: *Theory of Graphs* (Erdos and Katona, eds), Proceedings of the Colloquium held at Tihany (Hungary), September 1966, North-Holland, Amsterdam, 1968, pp. 219–230.
9. M. Las Vergnas, On the evaluation $t(G; 3, 3)$ of the Tutte polynomial of a graph, submitted.
10. P. Martin, Remarkable valuation of the dichromatic polynomial of planar multigraphs, *J. Combin. Theory, Ser. B* **24** (1978), 318–324.
11. P. Rosenstiehl and R. C. Read, On the principal edge tripartition of a graph, *Ann. Discrete Math.* **3** (1978), 195–226.
12. W. T. Tutte, Connectivity in matroids, *Canad. J. Math.* **18** (1966), 1301–1324.
13. D. J. A. Welsh, *Matroid Theory*, Academic Press, London, 1976.
14. A. Bouchet, Deficiency polynomials and orienting vectors of isotropic systems (submitted).
15. A. Bouchet, Connectivity of isotropic systems, *Proceedings of the New York International Conference in Combinatorics*, 1985 (to appear).

*References [14–19] were added in proof and so are not in alphabetic order.

16. A. Bouchet, Reducing prime graphs and recognizing circle graphs, *Combinatorica* (to appear).
17. A. Bouchet, Digraph decompositions and eulerian systems, *SIAM Journal on Algebraic and Discrete Methods* (to appear).
18. A. Bouchet, Transforming trees by local complementations. *Journal of Graph Theory* (to appear).
19. M. Las Vergnas, Le polynôme de Martin d'un graphe eulérien, *Ann. Disc. Math.* **17** (1983), 397–411.

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